

WPIlib Example Projects / WPIlib 示例项目

Warning / 警告

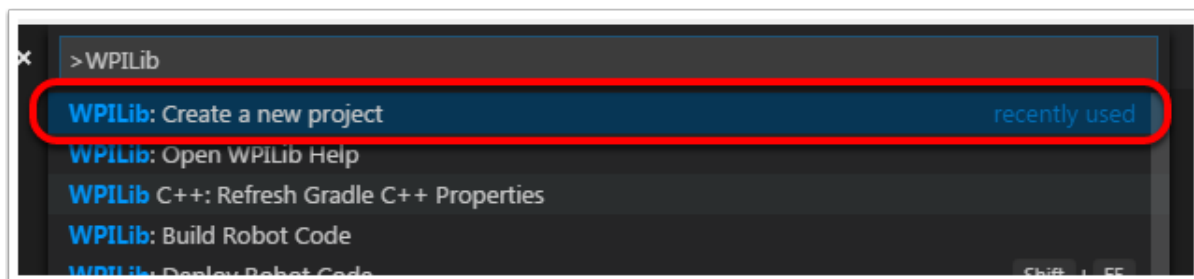
While every attempt is made to keep WPIlib examples functional, they are not intended to be used "as-is." At the very least, robot-specific constants will need to be changed for the code to work on a user robot. Many empirical constants have their values "faked" for demonstration purposes. Users are strongly encouraged to write their own code (from scratch or from an existing template) rather than copy example code.

虽然 WPIlib 会尽力保持示例可用，但这些示例并不适合“原样”使用。至少需要修改机器人专属常量，代码才能在用户自己的机器人上运行。许多经验常量的数值是为了演示而“虚构”的。强烈建议用户自行编写代码（从零开始或基于现有模板），而不是复制示例代码。

WPIlib example projects demonstrate a large number of library features and use patterns. Projects range from simple demonstrations of a single functionality to complete, competition-capable robot programs. All of these examples are available in VS Code by entering Ctrl+Shift+P, then selecting WPIlib: Create a new project and choosing example.

WPIlib

示例项目展示了大量库功能和使用模式。这些项目既有针对单一功能的简单演示，也有完整且具备比赛能力的机器人程序。在 VS Code 中按 Ctrl+Shift+P，选择 WPIlib: Create a new project，再选择示例，即可找到所有这些项目。



Basic Examples / 基础示例

These examples demonstrate basic/minimal robot functionality. They are useful for beginning teams who are gaining initial familiarity with robot programming, but are highly limited in functionality.

这些示例演示基础或最简的机器人功能。它们适合刚开始熟悉机器人编程的队伍，但功能非常有限。

- [Arcade Drive \(Java\)](#): Demonstrates a simple differential drive implementation using "arcade"-style controls through the DifferentialDrive class.

[Arcade Drive \(Java\)](#)：演示通过 DifferentialDrive 类，使用“街机式”操控实现简单的差速驱动。

- [Arcade Drive Xbox Controller \(Java\)](#): Demonstrates the same functionality seen in the previous example, except using an XboxController instead of an ordinary joystick.

[Arcade Drive Xbox Controller \(Java\)](#)：演示与上一个示例相同的功能，但使用 XboxController 代替普通摇杆。

- [Getting Started \(Java\)](#): Demonstrates a simple autonomous routine that drives forwards for two seconds at half speed.

[Getting Started \(Java\)](#)：演示一个简单的自动程序：以半速向前行驶两秒。

- [Mecanum Drive \(Java\)](#): Demonstrates a simple mecanum drive implementation using the MecanumDrive class.

[Mecanum Drive \(Java\)](#)：演示通过 MecanumDrive 类实现简单的麦克纳姆轮驱动。

- [Motor Controller \(Java\)](#): Demonstrates how to control the output of a motor with a joystick with an encoder to read motor position.

[Motor Controller \(Java\)](#)：演示如何使用摇杆控制电机输出，并通过编码器读取电机位置。

- [Simple Vision \(Java\)](#): Demonstrates how to stream video from a USB camera to the dashboard.

[Simple Vision \(Java\)](#)：演示如何将 USB 摄像头的视频流传输到仪表板。

- [Relay \(Java\)](#): Demonstrates the use of the Relay class to control a relay output with a set of joystick buttons.

[Relay \(Java\)](#) : 演示如何使用 Relay 类，通过一组摇杆按钮控制继电器输出。

- [Solenoids \(Java\)](#): Demonstrates the use of the Solenoid and DoubleSolenoid classes to control solenoid outputs with a set of joystick buttons.

[Solenoids \(Java\)](#) : 演示如何使用 Solenoid 和 DoubleSolenoid 类，通过一组摇杆按钮控制电磁阀输出。

- [TankDrive \(Java\)](#): Demonstrates a simple differential drive implementation using "tank"-style controls through the DifferentialDrive class.

[TankDrive \(Java\)](#) : 演示通过 DifferentialDrive 类，使用“坦克式”操控实现简单的差速驱动。

- [Tank Drive Xbox Controller \(Java\)](#): Demonstrates the same functionality seen in the previous example, except using an XboxController instead of an ordinary joystick.

[Tank Drive Xbox Controller \(Java\)](#) : 演示与上一个示例相同的功能，但使用 XboxController 代替普通摇杆。

Control Examples / 控制示例

These examples demonstrate WPIlib implementations of common robot controls. Sensors may be present, but are not the emphasized concept of these examples.

这些示例演示 WPIlib 对常见机器人控制功能的实现。示例中可能包含传感器，但传感器不是其重点。

- [DifferentialDriveBot \(Java\)](#): Demonstrates an advanced differential drive implementation, including encoder-and-gyro odometry through the DifferentialDriveOdometry class, and composition with PID velocity control through the DifferentialDriveKinematics and PIDController classes.

[DifferentialDriveBot \(Java\)](#) : 演示一种高级差速驱动实现：通过 DifferentialDriveOdometry 类，结合编码器和陀螺仪进行里程计 (odometry) 计算；并通过 DifferentialDriveKinematics 和 PIDController 类组合实现 PID 速度控制。

- [DifferentialDrivePoseEstimator \(Java\)](#): Demonstrates an advanced differential drive implementation with all the features of DifferentialDriveBot. In addition this example uses DifferentialDrivePoseEstimator to track the robots position on the field. It demonstrates these features by using WPIlib Simulation.

[DifferentialDrivePoseEstimator \(Java\)](#) : 演示一种高级差速驱动实现，包含 DifferentialDriveBot 的全部功能。此外，该示例使用 DifferentialDrivePoseEstimator 跟踪机器人在场地上的位置，并使用 WPIlib 仿真来演示这些功能。

- [Elevator with Profiled PID Controller \(Java\)](#): Demonstrates the use of the ProfiledPIDController class to control the position of an elevator mechanism.

[Elevator with Profiled PID Controller \(Java\)](#) : 演示如何使用 ProfiledPIDController 类控制升降机构的位置。

- [Elevator with Trapezoid Profiled PID \(Java\)](#): Demonstrates the use of the TrapezoidProfile class in conjunction with a "smart motor controller" to control the position of an elevator mechanism.

[Elevator with Trapezoid Profiled PID \(Java\)](#): 演示如何结合 TrapezoidProfile 类与“智能电机控制器”，控制升降机构的位置。

- [Elevator with Exponential Profiled PID \(Java\)](#): Demonstrates the use of the ExponentialProfile class in conjunction with a "smart motor controller" to control the position of an elevator mechanism.

[Elevator with Exponential Profiled PID \(Java\)](#) : 演示如何结合 ExponentialProfile 类与“智能电机控制器”，控制升降机构的位置。

- [Flywheel Bang-Bang Controller \(Java\)](#): Uses the BangBangController class to simply yet effectively control a flywheel.

[Flywheel Bang-Bang Controller \(Java\)](#) : 使用 BangBangController 类，以简单而有效的方式控制飞轮。

- [Gyro Mecanum \(Java\)](#): Demonstrates field-oriented control of a mecanum robot through the MecanumDrive class in conjunction with a gyro.

[Gyro Mecanum \(Java\)](#) : 演示如何结合陀螺仪，通过 MecanumDrive 类对麦克纳姆轮机器人进行场地坐标系控制。

- [MecanumBot](#) (Java): Demonstrates an advanced mecanum drive implementation, including encoder-and-gyro odometry through the MecanumDriveOdometry class, and composition with PID velocity control through the MecanumDriveKinematics and PIDController classes.

[MecanumBot](#) (Java) : 演示一种高级麦克纳姆轮驱动实现: 通过 MecanumDriveOdometry 类, 结合编码器和陀螺仪进行里程计计算; 并通过 MecanumDriveKinematics 和 PIDController 类组合实现 PID 速度控制。

- [Mecanum Drive PoseEstimator](#) (Java): Demonstrates an advanced mecanum drive implementation with all the features of MecanumBot. In addition this example uses MecanumDrivePoseEstimator to track the robots position on the field.

[Mecanum Drive PoseEstimator](#) (Java) : 演示一种高级麦克纳姆轮驱动实现, 包含 MecanumBot 的全部功能。此外, 该示例使用 MecanumDrivePoseEstimator 跟踪机器人在场地上的位置。

- [PotentiometerPID](#) (Java): Demonstrates the use of the PIDController class and a potentiometer to control the position of an elevator mechanism.

[PotentiometerPID](#) (Java) : 演示如何使用 PIDController 类和电位器控制升降机构的位置。

- [SwerveBot](#) (Java): Demonstrates an advanced swerve drive implementation, including encoder-and-gyro odometry through the SwerveDriveOdometry class, and composition with PID position and velocity control through the SwerveDriveKinematics and PIDController classes.

[SwerveBot](#) (Java) : 演示一种高级 swerve drive 实现: 通过 SwerveDriveOdometry 类, 结合编码器和陀螺仪进行里程计计算; 并通过 SwerveDriveKinematics 和 PIDController 类组合实现 PID 位置与速度控制。

- [Swerve Drive PoseEstimator](#) (Java): Demonstrates an advanced swerve drive implementation with all the features of SwerveBot. In addition this example uses SwerveDrivePoseEstimator to track the robots position on the field.

[Swerve Drive PoseEstimator](#) (Java) : 演示一种高级 swerve drive 实现, 包含 SwerveBot 的全部功能。此外, 该示例使用 SwerveDrivePoseEstimator 跟踪机器人在场地上的位置。

- [UltrasonicPID](#) (Java): Demonstrates the use of the PIDController class in conjunction with an ultrasonic sensor to drive to a set distance from an object.

[UltrasonicPID](#) (Java) : 演示如何结合 PIDController 类与超声波传感器, 使机器人行驶到距物体设定的距离。

Sensor Examples / 传感器示例

These examples demonstrate sensor reading and data processing using WPILib. Mechanisms control may be present, but is not the emphasized concept of these examples.

这些示例演示使用 WPILib 读取传感器和处理数据。示例中可能包含机构控制, 但机构控制不是其重点。

- [HTTP Camera](#) (Java): Demonstrates the use of OpenCV and a HTTP Camera to overlay a rectangle on a captured video feed and stream it to the dashboard.

[HTTP Camera](#) (Java) : 演示如何使用 OpenCV 和 HTTP 摄像头, 在采集的视频画面上叠加矩形, 并将视频流传输到仪表板。

- [Power Distribution CAN Monitoring](#) (Java): Demonstrates obtaining sensor information from a Power Distribution module over CAN using the PowerDistribution class.

[Power Distribution CAN Monitoring](#) (Java) : 演示如何使用 PowerDistribution 类, 通过 CAN 从配电模块获取传感器信息。

- [Duty Cycle Encoder](#) (Java): Demonstrates the use of the DutyCycleEncoder class to read values from a PWM-type absolute encoder.

[Duty Cycle Encoder](#) (Java) : 演示如何使用 DutyCycleEncoder 类读取 PWM 型绝对编码器的数值。

- [DutyCycleInput](#) (Java): Demonstrates the use of the DutyCycleInput class to read the frequency and fractional duty cycle of a PWM input.

[DutyCycleInput](#) (Java) : 演示如何使用 DutyCycleInput 类读取 PWM 输入的频率和以比例表示的占空比。

- [Encoder](#) (Java): Demonstrates the use of the Encoder class to read values from a quadrature encoder.

[Encoder](#) (Java) : 演示如何使用 Encoder 类读取正交编码器的数值。

- [Gyro](#) (Java): Demonstrates the use of the AnalogGyro class to measure robot heading and stabilize driving.

[Gyro](#) (Java) : 演示如何使用 AnalogGyro 类测量机器人朝向并稳定驾驶控制。

- [Intermediate Vision](#) (Java): Demonstrates the use of OpenCV and a USB camera to overlay a rectangle on a captured video feed and stream it to the dashboard.

[Intermediate Vision](#) (Java) : 演示如何使用 OpenCV 和 USB 摄像头, 在采集的视频画面上叠加矩形, 并将视频流传输到仪表板。

- [AprilTagsVision](#) (Java): Demonstrates on-roboRIO detection of AprilTags using an attached USB camera.

[AprilTagsVision](#) (Java) : 演示如何使用连接的 USB 摄像头, 在 roboRIO 上检测 AprilTag。

- [Ultrasonic](#) (Java): Demonstrates the use of the Ultrasonic class to read data from an ultrasonic sensor in conjunction with the MedianFilter class to reduce signal noise.

[Ultrasonic](#) (Java) : 演示如何使用 Ultrasonic 类读取超声波传感器的数据, 并结合 MedianFilter 类降低信号噪声。

- [SysIdRoutine](#) (Java): Demonstrates the use of the SysIdRoutine API to gather characterization data for a differential drivetrain.

[SysIdRoutine](#) (Java) : 演示如何使用 SysIdRoutine API 采集差速底盘的特性分析数据。

Command-Based Examples / Command-Based 示例

These examples demonstrate the use of the [Command-Based framework](#).

这些示例演示 [Command-Based](#) 框架的用法。

- [DriveDistanceOffboard](#) (Java): Demonstrates the use of a TrapezoidProfileCommand in conjunction with a "smart motor controller" to drive forward by a set distance with a trapezoidal motion profile.

[DriveDistanceOffboard](#) (Java) : 演示如何结合 TrapezoidProfileCommand 与“智能电机控制器”, 按照梯形运动曲线向前行驶设定的距离。

- [Rapid React Command Bot](#) (Java): This project uses the latest command based best practices and the Epilogue logging system. It is capable of playing the FRC 2022 game Rapid React.

[Rapid React Command Bot](#) (Java) : 该项目采用最新的 Command-Based 最佳实践和 Epilogue 日志系统, 能够参与 FRC 2022 赛季的 Rapid React 比赛。

- [Inlined Hatchbot](#) (Java): A complete set of robot code for a simple hatch-delivery bot typical of the 2019 FRC game Destination: Deep Space. Commands are written in an "inline" style, in which explicit subclassing of Command is avoided.

[Inlined Hatchbot](#) (Java) : 一套完整的机器人代码, 适用于 2019 年 FRC 比赛 Destination: Deep Space 中典型的简易舱门板运送机器人。Command 采用“内联”风格编写, 避免显式继承 Command。

- [Traditional Hatchbot](#) (Java): A complete set of robot code for a simple hatch-delivery bot typical of the 2019 FRC game Destination: Deep Space. Commands are written in a "traditional" style, in which subclasses of Command are written for each robot action.

[Traditional Hatchbot](#) (Java) : 一套完整的机器人代码, 适用于 2019 年 FRC 比赛 Destination: Deep Space 中典型的简易舱门板运送机器人。Command 采用“传统”风格编写, 为机器人的每个动作编写 Command 子类。

- [MecanumControllerCommand](#) (Java): Demonstrates trajectory generation and following with a mecanum drive using the TrajectoryGenerator and MecanumControllerCommand classes.

[MecanumControllerCommand](#) (Java) : 演示如何使用 TrajectoryGenerator 和 MecanumControllerCommand 类, 为麦克纳姆轮驱动生成并跟踪轨迹。

- [Select Command Example](#) (Java): Demonstrates the use of the SelectCommand class to run one of a selection of commands depending on a runtime-evaluated condition.

[Select Command Example](#) (Java) : 演示如何使用 SelectCommand 类, 根据运行时求值的条件, 从一组选项中选择并运行一个 Command。

- [SwerveControllerCommand](#) (Java): Demonstrates trajectory generation and following with a swerve drive using the TrajectoryGenerator and SwerveControllerCommand classes.

[SwerveControllerCommand](#) (Java) : 演示如何使用 TrajectoryGenerator 和 SwerveControllerCommand 类, 为 swerve drive 生成并跟踪轨迹。

State-Space Examples / 状态空间示例

These examples demonstrate the use of the [State-Space Control](#).

这些示例演示状态空间控制的用法。

- [StateSpaceFlywheel](#) (Java): Demonstrates state-space control of a flywheel.

[StateSpaceFlywheel](#) (Java) : 演示飞轮的状态空间控制。

- [StateSpaceFlywheelSysId](#) (Java): Demonstrates state-space control using SysId's System Identification for controlling a flywheel.

[StateSpaceFlywheelSysId](#) (Java) : 演示如何使用 SysId 的辨识结果, 对飞轮进行状态空间控制。

- [StateSpaceElevator](#) (Java): Demonstrates state-space control of an elevator.

[StateSpaceElevator](#) (Java) : 演示升降机构的状态空间控制。

- [StateSpaceArm](#) (Java): Demonstrates state-space control of an Arm.

[StateSpaceArm](#) (Java) : 演示机械臂的状态空间控制。

Simulation Physics Examples / 物理仿真示例

These examples demonstrate the use of the physics simulation.

这些示例演示物理仿真的用法。

- [ElevatorSimulation](#) (Java): Demonstrates the use of physics simulation with a simple elevator.

[ElevatorSimulation](#) (Java) : 演示如何对简单的升降机构使用物理仿真。

- [ElevatorSimulation with Exponential PID](#) (Java): Demonstrates the use of physics simulation of an elevator being controlled with exponential profiled PID.

[ElevatorSimulation with Exponential PID](#) (Java) : 演示如何对使用指数运动曲线 PID 控制的升降机构进行物理仿真。

- [ArmSimulation](#) (Java): Demonstrates the use of physics simulation with a simple single-jointed arm.

[ArmSimulation](#) (Java) : 演示如何对简单的单关节机械臂使用物理仿真。

- [SimpleDifferentialDriveSimulation](#) (Java): A barebones example of a basic drivetrain that can be used in simulation.

[SimpleDifferentialDriveSimulation](#) (Java) : 一个可用于仿真的基础底盘最简示例。

Miscellaneous Examples / 其他示例

These examples demonstrate miscellaneous WPILib functionality that does not fit into any of the above categories.

这些示例演示不属于上述类别的其他 WPILib 功能。

- [Addressable LED](#) (Java): Demonstrates the use of the AddressableLED class to control RGB LEDs for robot decoration and/or driver feedback.

[Addressable LED](#) (Java) : 演示如何使用 AddressableLED 类控制 RGB LED, 用于机器人装饰和/或向操作手提供反馈。

- [Digital Communication](#) (Java): This is a sample program demonstrating how to communicate to a light controller from the robot code using the roboRIO's DIO ports.

[Digital Communication](#) (Java) : 此示例程序演示如何通过 roboRIO 的 DIO 端口, 在机器人代码中与灯光控制器通信。

- [I2C Communication](#) (Java): This is a sample program demonstrating how to communicate to a light controller from the robot code using the roboRIO's I2C port.

[I2C Communication](#) (Java) : 此示例程序演示如何通过 roboRIO 的 I2C 端口, 在机器人代码中与灯光控制器通信。

- [DMA](#) (Java): Demonstrates the use of DMA (Direct Memory Access) to read from sensors without using the RoboRIO's CPU.

[DMA](#) (Java) : 演示如何使用 DMA (直接内存访问) 从传感器读取数据, 而不使用 roboRIO 的 CPU。

- [HID Rumble](#) (Java): Demonstrates the use of the "rumble" functionality for tactile feedback on supported HID's (such as XboxControllers).

[HID Rumble](#) (Java) : 演示如何在支持此功能的 HID (例如 XboxController) 上使用“振动”功能提供触觉反馈。

- [Shuffleboard](#) (Java): Demonstrates configuring tab/widget layouts on the "Shuffleboard" dashboard from robot code through the Shuffleboard class's fluent builder API.

[Shuffleboard](#) (Java) : 演示如何在机器人代码中，通过 [Shuffleboard](#) 类的流式构建器 API，配置 [Shuffleboard](#) 仪表板的选项卡和组件布局。

- [RomiReference](#) (Java): A command based example of how to run the [Romi robot](#).

[RomiReference](#) (Java) : 一个 Command-Based 示例，演示如何运行 [Romi 机器人](#)。

- [XRPRReference](#) (Java): A command based example of how to run the [XRP robot](#).

[XRPRReference](#) (Java) : 一个 Command-Based 示例，演示如何运行 [XRP 机器人](#)。

- [Mechanism2d](#) (Java): A simple example of using [Mechanism2d](#).

[Mechanism2d](#) (Java) : 一个使用 [Mechanism2d](#) 的简单示例。

- [EventLoop](#) (Java): Demonstrates the use of the EventLoop class that allows code to be called based on a boolean condition in the event-driven programming style.

[EventLoop](#) (Java) : 演示如何使用 [EventLoop](#) 类，以事件驱动编程方式根据布尔条件调用代码。

- [UnitTest](#) (Java): Shows how to do [Unit Testing](#). The test files need to be in a separate Test directory ([Java](#)).

[UnitTest](#) (Java) : 演示如何进行[单元测试](#)。测试文件需要放在单独的 Test 目录中 ([Java](#))。